Chapter ML:I

I. Introduction

- Examples of Learning Tasks
- Specification of Learning Tasks
- Elements of Machine Learning
- Comparative Syntax Overview
- Classification Approaches

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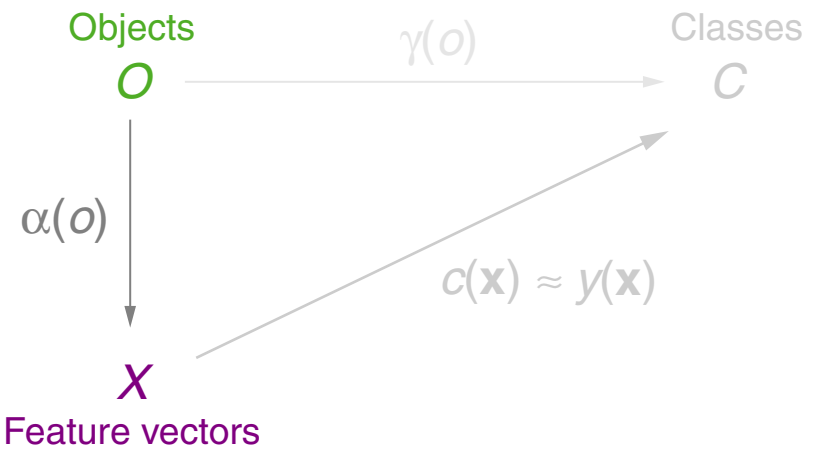
Elements of Machine Learning

(1) Model Formation: Real World → Model World



Related questions:

- From what kind of experience should be learned?
- Which level of fidelity is sufficient to solve a certain task?

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(2) Construction of Model Function

Optimization approach

Optimization objective
Loss function [+ Regularization]

Model function $\rightsquigarrow$ Hypothesis space

Task

Data

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(2) Construction of Model Function: LMS in a Nutshell

Optimization approach

Optimization objective
Loss function [+ Regularization]

Model function $\rightsquigarrow$ Hypothesis space

Task

Binary classification

Data

$$D = \{(\mathbf{x}_1, c(\mathbf{x}_1)), \ldots, (\mathbf{x}_n, c(\mathbf{x}_n))\} \subseteq X \times \{-1, 1\}$$

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(2) Construction of Model Function: LMS in a Nutshell (continued)

Optimization approach

Optimization objective
Loss function [+ Regularization]

Model function $\rightsquigarrow$ Hypothesis space

- Hypothesis space: $\mathbf{w} \in \mathbf{R}^{p+1}$
- Linear model: $y(\mathbf{x}) = w_0 + \sum_{i=1}^{p} w_i x_i$

Task

Binary classification

Data

$D = \{(\mathbf{x}_1, c(\mathbf{x}_1)), \ldots, (\mathbf{x}_n, c(\mathbf{x}_n))\} \subseteq X \times \{-1, 1\}$

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(2) Construction of Model Function: LMS in a Nutshell (continued)

Optimization approach	
Optimization objective Loss function [+ Regularization]	❑ Objective: Minimize squared loss ❑ Regularization: None ❑ Loss: Sum of squared residuals
Model function $\rightsquigarrow$ Hypothesis space	❑ Hypothesis space: $\mathbf{w} \in \mathbf{R}^{p+1}$ ❑ Linear model: $y(\mathbf{x}) = w_0 + \sum_{i=1}^{p} w_i x_i$
Task	Binary classification
Data	$D = \{(\mathbf{x}_1, c(\mathbf{x}_1)), \ldots, (\mathbf{x}_n, c(\mathbf{x}_n))\} \subseteq X \times \{-1, 1\}$

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(2) Construction of Model Function: LMS in a Nutshell (continued)

Optimization approach	Stochastic gradient descent
Optimization objective Loss function [+ Regularization]	❑ Objective: Minimize squared loss ❑ Regularization: None ❑ Loss: Sum of squared residuals
Model function $\rightsquigarrow$ Hypothesis space	❑ Hypothesis space: $\mathbf{w} \in \mathbf{R}^{p+1}$ ❑ Linear model: $y(\mathbf{x}) = w_0 + \sum_{i=1}^{p} w_i x_i$
Task	Binary classification
Data	$D = \{(\mathbf{x}_1, c(\mathbf{x}_1)), \ldots, (\mathbf{x}_n, c(\mathbf{x}_n))\} \subseteq X \times \{-1, 1\}$

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Related questions:

- What are useful classes of model functions?
- What are methods to fit (= learn) model functions?
- What are measures to assess the goodness of fit?

- How does (label) noise affect the learning process?
- How does the example number affect the learning process?
- How to deal with extreme class imbalance?

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(3) Feature Space Structure

The feature space is an inner product space.

- An inner product space (also called pre-Hilbert space) is a vector space with an additional structure called “inner product”.
- Example: Euclidean vector space equipped with the dot product.
- Enables algorithms such as gradient descent and support vector machines.

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The feature space is a σ-algebra.

- A σ-algebra on a set X is a collection of subsets of X that includes X itself, is closed under complement, and is closed under countable unions.
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The feature space is a finite set of vectors with nominal dimensions.

- Requires concept learning via set splitting as done by decision trees.

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Remarks:

- The aforementioned examples of feature spaces are not meant to be complete. However, they illustrate a broad range of structures underlying the example sets we want to learn from.
- The structure of a feature space constrains the applicable learning algorithm. Usually, this structure is inherently determined by the application domain and cannot be chosen.

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(4) Discriminative vs. Generative Approach to Classification Problems

- Discriminative classifiers (models) learn a boundary between classes.
- Generative classifiers exploit the distributions underlying the classes.

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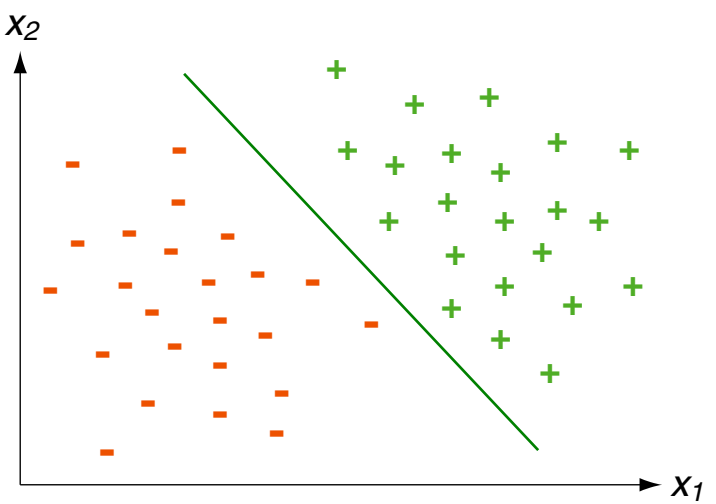
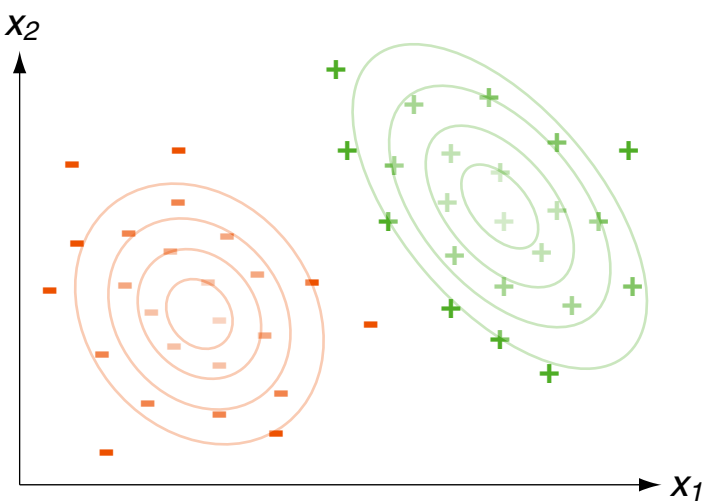


discriminative
$\rightsquigarrow$ classification rule



generative
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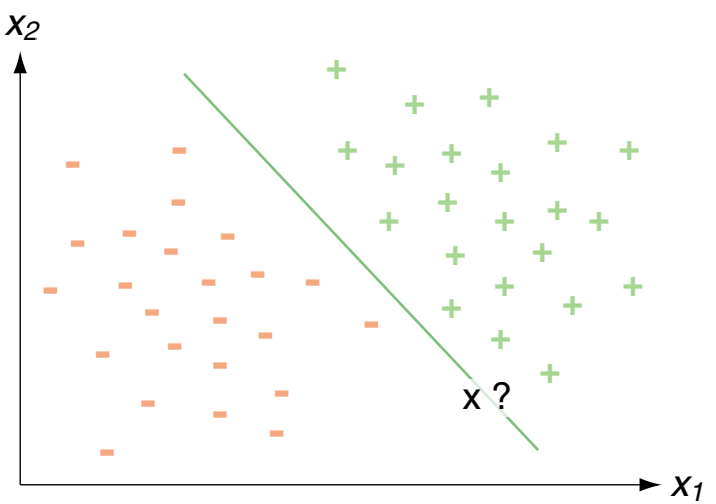

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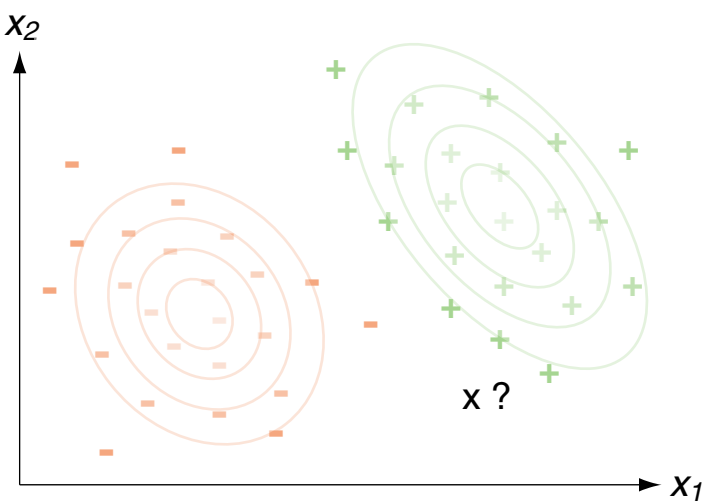

generative
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Remarks:

- When classifying a new example (1) discriminative classifiers apply a decision rule that was learned via minimizing the misclassification rate given training examples D, while (2) generative classifiers maximize the probability of the combined event $P(\mathbf{x}, y)$, or, similarly, the a-posteriori probability $P(y \mid \mathbf{x})$, $y \in \{\ominus, \oplus\}$.
- The LMS algorithm computes "only" a decision boundary, i.e., it constructs a discriminative classifier. A Bayes classifier is an example for a generative model.
- Yoav Freund provides an excellent video illustrating the pros and cons of discriminative and generative models respectively. [YouTube]
- Discriminative models may be further differentiated in models that also determine the posterior class probabilities $P(y \mid \mathbf{x})$ (without computing the joint probabilities $P(\mathbf{x}, y)$) and those that do not. In the latter case, only a so-called "discriminant function" is computed.

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Elements of Machine Learning

(5) Frequentist vs. Subjectivist Approach to Parameter Estimation

Frequentism:

- There is a (hidden) mechanism that generates D.
- To model this mechanism you consider
  - a family of distributions, or
  - a model function, or
  - a combination of both,

  parameterized by θ. The possible values for θ form the hypothesis space H.
- Select a most probable hypothesis $h_{\text{ML}} \in H$ by estimating θ using a sample $D' \subset D$. h_{ML} is called maximum likelihood hypothesis.

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$$\theta \rightsquigarrow D', \qquad h_{\text{ML}} = \operatorname*{argmax}_{h \in H} P(D' \mid h)$$

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Remarks:

- θ is a parameter or a parameter vector that is considered as fixed (in particular: not as a random variable), but unknown.
- In the experiment of flipping a coin, one may suppose a Laplace experiment and consider the binomial distribution, $B(n, p)$.
- $P(D' \mid h)$ is the probability of observing D' under h. I.e., it is the probability of observing D' if the hidden mechanism that generates D' behaves according to the considered model whose parameter θ is set to h.

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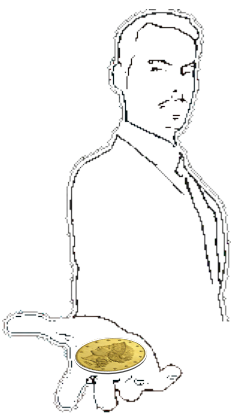
(5) Frequentist vs. Subjectivist Approach to Parameter Estimation (continued)

Subjectivism:

- Consider a model for the mechanism that has generated D.
- There are different beliefs about the parameter (vector) θ that characterizes the model. The possible values for θ form the hypothesis space H.
- Select a most probable hypothesis $h_{\text{MAP}} \in H$ by weighting the ML estimates under D with the priors. h_{MAP} is called maximum a-posteriori hypothesis.

Belief/Prior 1: $P(\underbrace{p = 0.5}_{\theta_1})$ = 0.95



Belief/Prior 2: $P(\underbrace{p = 0.75}_{\theta_2})$ = 0.50

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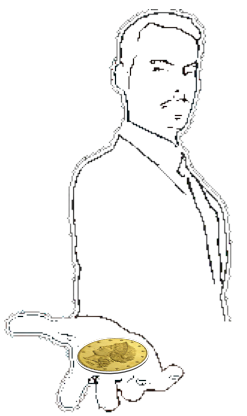

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Belief/Prior 2: $P(\underbrace{p = 0.75}_{\theta_2})$ = 0.50

$$\left.\begin{aligned} \theta_1 + D &\rightarrow P(D \mid \theta_1) \\ \theta_2 + D &\rightarrow P(D \mid \theta_2) \end{aligned}\right\}$$

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(5) Frequentist vs. Subjectivist Approach to Parameter Estimation (continued)

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$$\left.\begin{array}{l} \theta_1 + D \;\rightarrow\; P(D \mid \theta_1) \\ \theta_2 + D \;\rightarrow\; P(D \mid \theta_2) \end{array}\right\} \quad h_{\text{MAP}} = \underset{h \in \{\theta_1, \theta_2\}}{\text{argmax}}\, P(h \mid D) = \underset{h \in \{\theta_1, \theta_2\}}{\text{argmax}} \frac{P(D \mid h) \cdot P(h)}{P(D)}$$

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Remarks:

- θ is considered as random variable. There is prior knowledge about the distribution of θ.
- p is a parameter of the binomial distribution and denotes the success probability for each trial.
  - Belief 1: With a probability of 0.95 the coin is fair (both sides are equally likely).
  - Belief 2: With a probability of 0.5 the odds of preferring a particular side is 3:1.

  Given D from a number of trials compute $P(D \mid \theta_1)$ and $P(D \mid \theta_2)$ and the respective values for $P(\theta_1 \mid D)$ and $P(\theta_2 \mid D)$.
- The subjectivist approach is also called Bayesian interpretation of probability.
- The Bayesian interpretation of probability enables by design the integration of prior knowledge, background knowledge, and human expertise.
- Food for thought: Discuss the use of frequentist and subjectivist approaches to decision making if you had to develop an AI that plays poker.

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